INCH-POUND
MIL-DTL-24704/2B(SH)
31 July 2019
SUPERSEDING
MIL-F-24704/2A(SH)
23 March 1992

DETAIL SPECIFICATION SHEET

FLANGES, FOUR-BOLT, SQUARE, BUTT WELD, FOR HYDRAULIC SYSTEMS

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24704.

Scope. This specification sheet covers the requirements for four-bolt square butt weld flanges for use in 3,000 pounds per square inch (lb/in^2) hydraulic systems.

Part or identifying number (PIN). PINs to be used for butt weld flanges acquired to this specification are created as follows:

M	24704	/2	X	X	XX	X	X
Specification prefix	Specification number	Specification sheet number	Flange style code (see table I)	Flange (square) size code (see table II)	Pipe and tube size code (see table III)	Material code (see table IV)	Vent requirement designation (see figures 5 and 6)

Example: The PIN for a butt weld straight flange for nominal pipe size (NPS) piping, flange size 4.25 inches square, NPS 2, made of type 316 corrosion-resistant steel (CRES), with a vent located at position X would be written as M24704/2SG16CX.

Flange style. The flange style designation letter indicates whether it's a straight or right angle flange and whether it's for tubing or NPS piping as specified in table I.

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TABLE I. Flange styles.

Code	Description
S	Straight flange for NPS piping (see figure 1)
L	Right angle flange for NPS piping (see figure 2)
T	Straight flange for outside diameter (OD) tubing (see figure 3)
R	Right angle flange for OD tubing (see figure 4)

Flange (square) size. The square size of the flange shall be designated by a letter as specified in table II (see figures 1, 2, 3, and 4 for listings of compatible flange and pipe and tube sizes).

TABLE II. Flange (square) sizes.

Code	Square size (inches)
A	1.50
B	1.75
C	2.00
D	2.25
E	2.75
F	3.50
G	4.25
H	5.75

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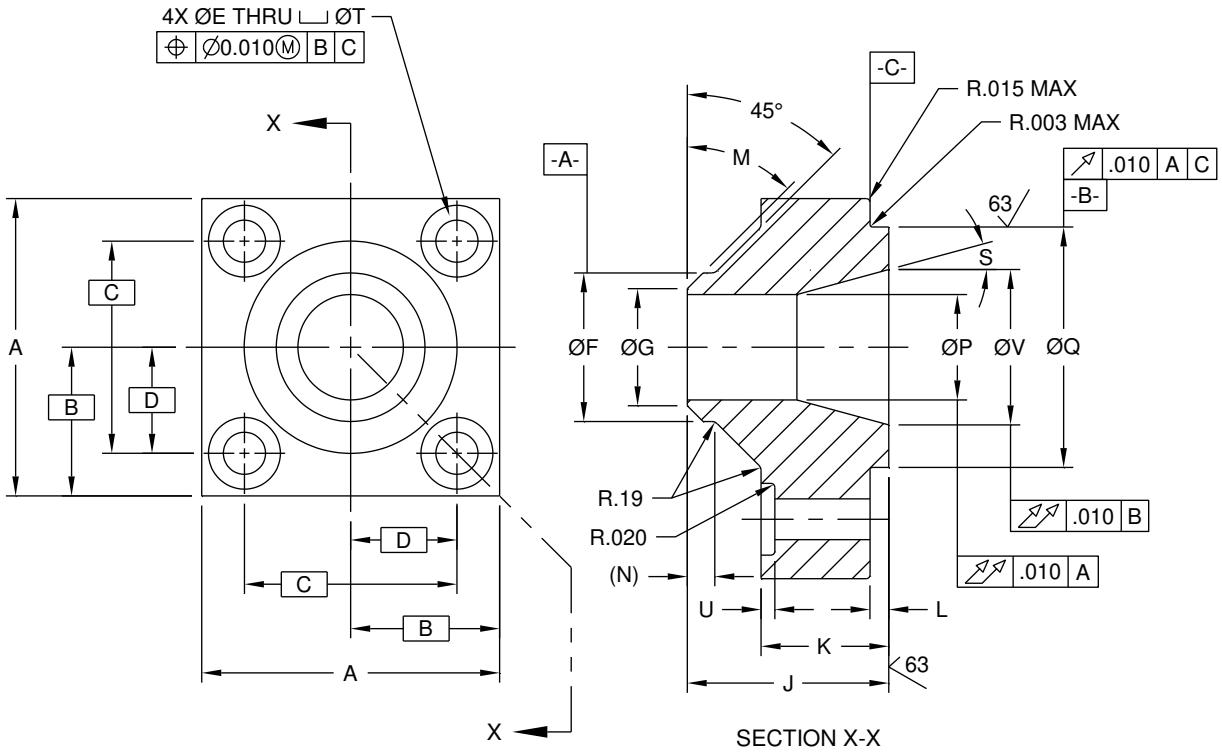


FIGURE 1. Straight flange – pipe.

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#	**	Flange size (inches)	Pipe size (NPS)	A (square)	B	C (square)	D	E ±0.010	F	G ±0.010	J	K
F	08	3.50	1.000	3.50	1.75	2.500	1.250	0.531	1.315	1.160	2.125	1.250
F	10	3.50	1.250	3.50	1.75	2.500	1.250	0.531	1.660	1.301	2.125	1.250
F	12	3.50	1.500	3.50	1.75	2.500	1.250	0.531	1.900	1.520	2.125	1.250
G	16	4.25	2.000	4.25	2.12	3.062	1.531	0.656	2.375	1.708	2.594	1.750
H	20	5.75	2.500	5.75	2.88	4.000	2.000	0.781	2.915	2.145	3.120	2.21

#	**	Flange size (inches)	Pipe size (NPS)	L +0.002 -0.000	M -0.000 +5.000 (degrees)	N (ref)	P +0.010 -0.000	Q +0.000 -0.002	V +0.000 -0.015	S ±2.5 (degrees)	T	U
F	08	3.50	1.000	0.125	40	0.375	0.957	2.124	1.500	15.0	0.915	0.093
F	10	3.50	1.250	0.125	40	0.375	1.281	2.124	1.500	5.0	0.915	0.218
F	12	3.50	1.500	0.125	40	0.375	1.500	2.124	-	-	0.915	0.218
G	16	4.25	2.000	0.125	40	0.375	1.688	2.374	1.750	15.0	1.228	0.500
H	20	5.75	2.500	0.125	40	0.438	2.125	2.999	-	-	-	-

NOTES:

1. Unless otherwise specified herein or in 6.2 of the base specification (MIL-DTL-24704):
   a. Surface roughness shall not exceed 125 microinches.
   b. Tolerances shall be:
      (1) X.XXX: ±0.015
      (2) X.XX: ±0.03
   c. Edges and corners shall be broken to a radius of 0.005 to 0.015 inch.
2. PIN --- M24704/2S#**xx (# and ** selected from above).

FIGURE 1. Straight flange – pipe – Continued.

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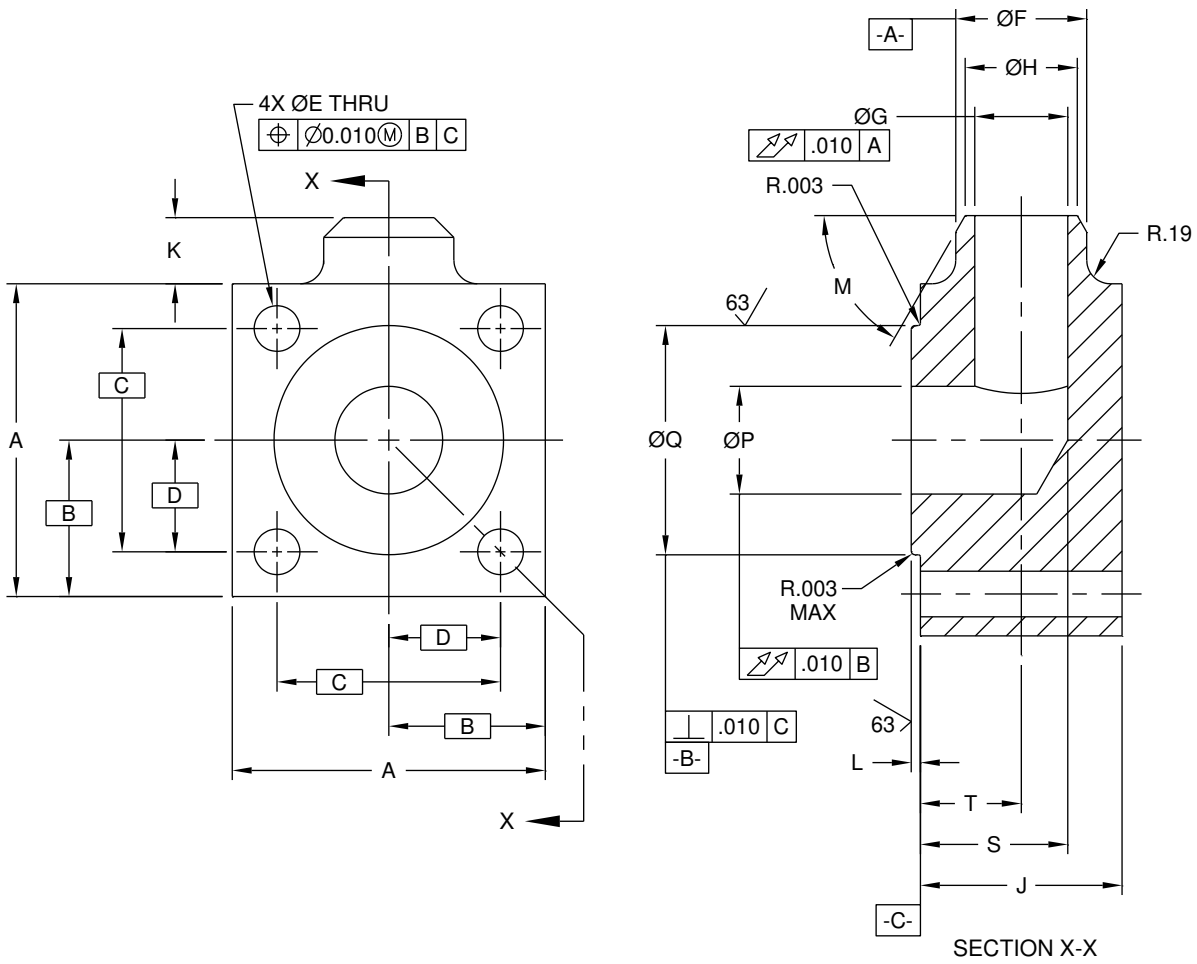


FIGURE 2. Right angle flange – pipe.

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#	**	Flange size (inches)	Pipe size (NPS)	A (square)	B	C (square)	D	E ±0.010	F	G ±0.010	H ±0.010	J	K
F	08	3.50	1.000	3.50	1.75	2.500	1.250	0.531	1.315	0.957	1.160	2.189	0.875
F	10	3.50	1.250	3.50	1.75	2.500	1.250	0.531	1.660	1.281	1.344	2.689	0.875
F	12	3.50	1.500	3.50	1.75	2.500	1.250	0.531	1.900	1.500	1.562	2.939	0.875
G	16	4.25	2.000	4.25	2.12	3.062	1.531	0.656	2.375	1.688	1.750	3.532	1.000

#	**	Flange size (inches)	Pipe size (NPS)	L +0.002 -0.000	M -0.000 +5.000 (degrees)	P +0.000 -0.015	Q +0.000 -0.002	T	S (maximum)
F	08	3.50	1.000	0.125	40	1.500	2.124	0.957	1.656
F	10	3.50	1.250	0.125	40	1.500	2.124	1.312	2.000
F	12	3.50	1.500	0.125	40	1.500	2.124	1.312	2.094
G	16	4.25	2.000	0.125	40	1.750	2.374	1.750	2.469

NOTES:

1. Unless otherwise specified herein or in 6.2 of the base specification (MIL-DTL-24704):
   a. Surface roughness shall not exceed 125 microinches.
   b. Tolerances shall be:
      (1) X.XXX: ±0.015
      (2) X.XX: ±0.03
   c. Edges and corners shall be broken to a radius of 0.005 inch to 0.015 inch.
2. PIN --- M24704/2L#**xx (# and ** selected from above).

FIGURE 2. Right angle flange – pipe – Continued.

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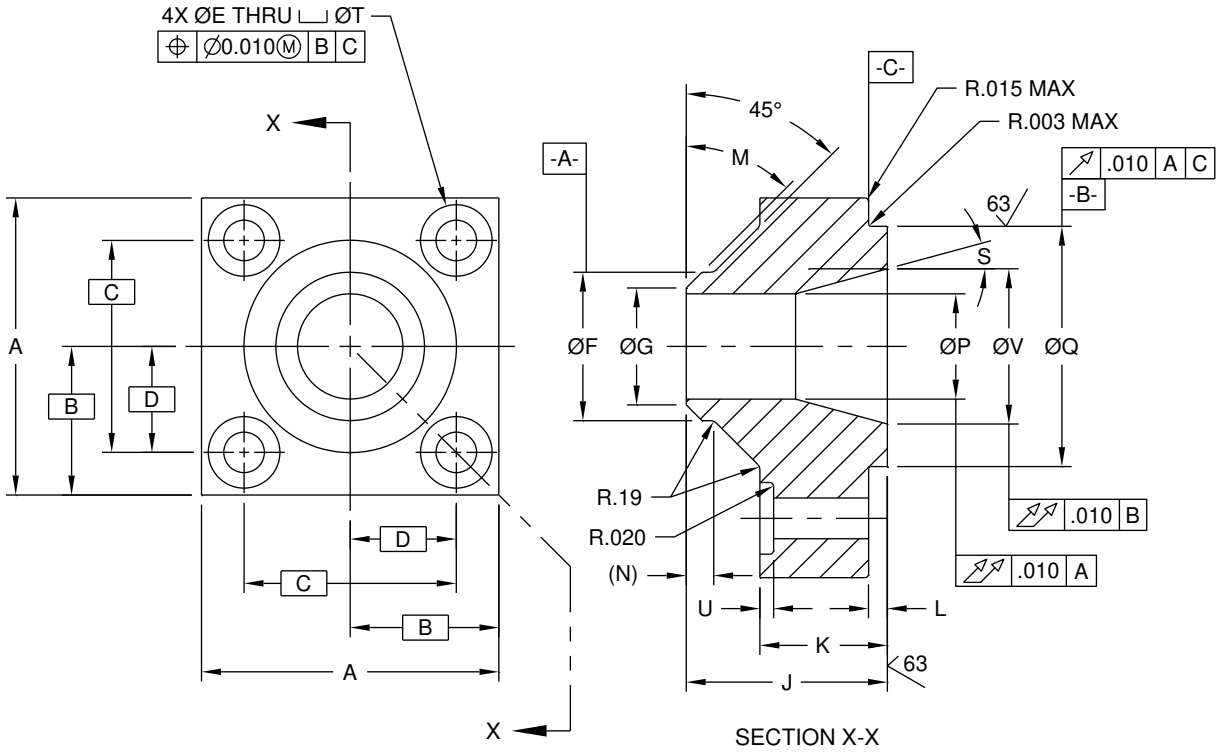


FIGURE 3. Straight flange – tube.

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#	**	Flange size (inches)	Tube size (OD)	A (square)	B	C (square)	D	E ±0.010	F	G ±0.010	J	K
F	10	3.50	1.250	3.50	1.75	2.500	1.250	0.531	1.250	0.748	2.125	1.250
F	12	3.50	1.500	3.50	1.75	2.500	1.250	0.531	1.500	0.872	2.125	1.250
G	16	4.25	2.000	4.25	2.12	3.062	1.531	0.656	2.000	1.185	2.594	1.750

#	**	Flange size (inches)	Tube size (OD)	L +0.002 -0.000	M -0.000 +5.000 (degrees)	N (ref)	P +0.010 -0.000	Q +0.000 -0.002	S ±2.5 (degrees)	T	U	V +0.000 -0.015
F	10	3.50	1.250	0.125	40	0.375	0.688	2.124	30.0	0.915	0.218	1.500
F	12	3.50	1.500	0.125	40	0.375	0.812	2.124	30.0	0.915	0.218	1.500
G	16	4.25	2.000	0.125	40	0.375	1.125	2.374	30.0	1.228	0.500	1.750

NOTES:

1. Unless otherwise specified herein or in 6.2 of the base specification (MIL-DTL-24704):
   a. Surface roughness shall not exceed 125 microinches.
   b. Tolerances shall be:
      (1) X.XXX: ±0.015
      (2) X.XX: ±0.03
   c. Edges and corners shall be broken to a radius of 0.005 inch to 0.015 inch.
2. PIN --- M24704/2T#**xx (# and ** selected from above).

FIGURE 3. Straight flange – tube – Continued.

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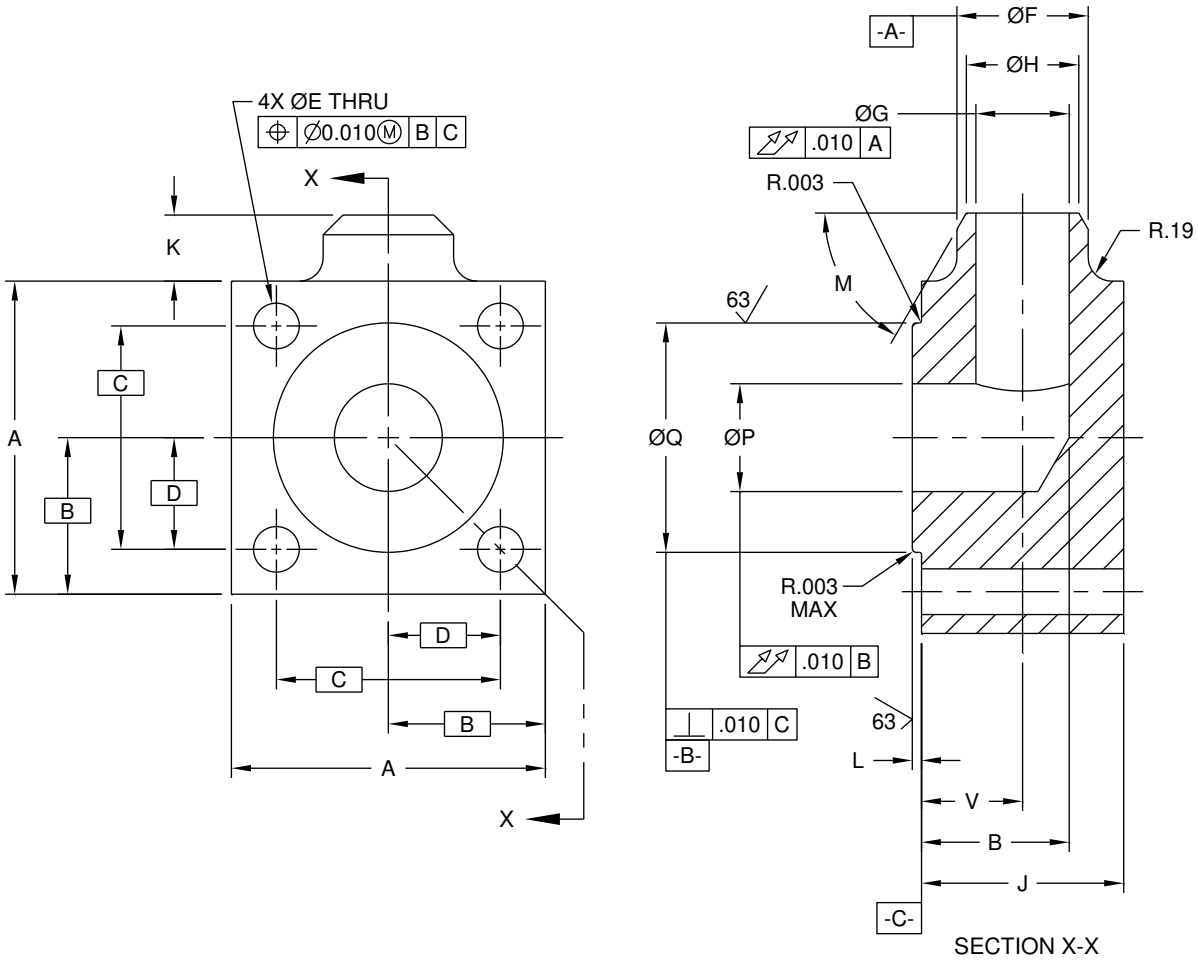


FIGURE 4. Right angle flange – tube.

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#	**	Flange size (inches)	Tube size (OD)	A (square)	B	C (square)	D	E ±0.010	F	G ±0.010	H ±0.010	J	K
F	10	3.50	1.250	3.50	1.75	2.500	1.250	0.531	1.250	0.688	0.748	2.689	0.875
F	12	3.50	1.500	3.50	1.75	2.500	1.250	0.531	1.500	0.812	0.872	2.939	0.875
G	16	4.25	2.000	4.25	2.12	3.062	1.531	0.656	2.000	1.125	1.185	3.532	1.000

#	**	Flange size (inches)	Tube size (OD)	L +0.002 -0.000	M -0.000 +5.000 (degrees)	P +0.000 -0.015	Q +0.000 -0.002	V	S (maximum)
F	10	3.50	1.250	0.125	40	0.688	2.124	1.312	1.656
F	12	3.50	1.500	0.125	40	0.812	2.124	1.312	1.718
G	16	4.25	2.000	0.125	40	1.500	2.374	1.750	2.312

NOTES:

1. Unless otherwise specified herein or in 6.2 of the base specification (MIL-DTL-24704):
   a. Surface roughness shall not exceed 125 microinches.
   b. Tolerances shall be:
      (1) X.XXX: ±0.015
      (2) X.XX: ±0.03
   c. Edges and corners shall be broken to a radius of 0.005 inch to 0.015 inch.
2. PIN --- M24704/2R#**xx (# and ** selected from above).

FIGURE 4. Right angle flange – tube – Continued.

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Pipe and tube size. The pipe or tubing size shall be designated as specified in table III.

TABLE III. Pipe and tube sizes.

Code	Pipe (NPS)	Tube (OD)
01	⅛ (0.405 OD)	--
02	¼ (0.540 OD)	0.250
03	⅜ (0.675 OD)	0.375
04	½ (0.840 OD)	0.500
06	¾ (1.050 OD)	0.750
08	1 (1.315 OD)	1.000
10	1¼ (1.660 OD)	1.250
12	1½ (1.900 OD)	1.500
16	2 (2.375 OD)	2.000
20	2½ (2.875 OD)	2.500

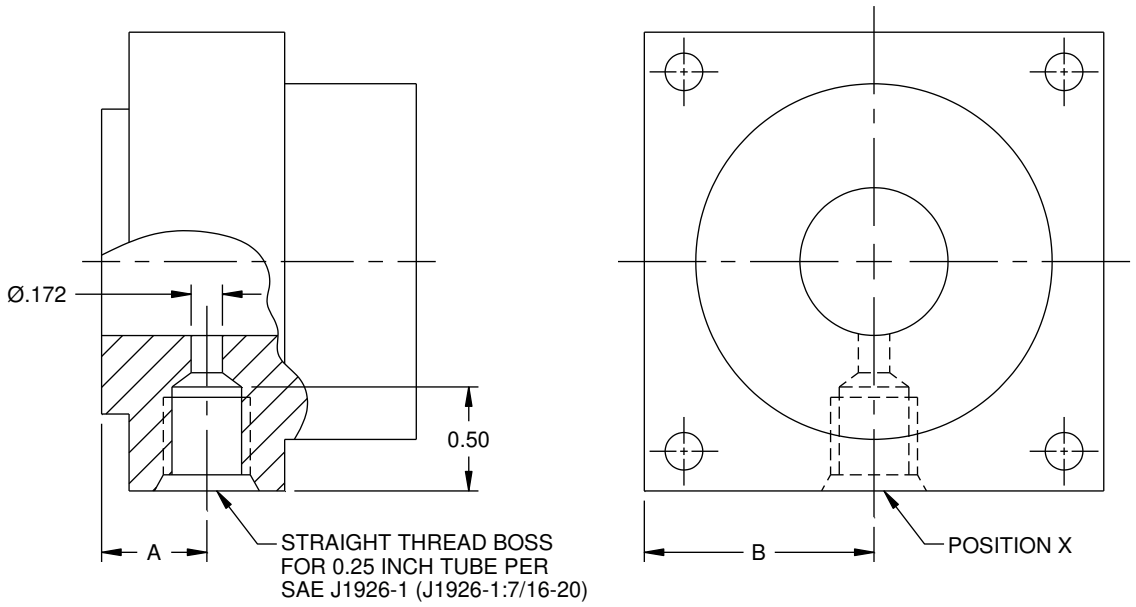
Material. The flange material shall be designated as specified in table IV.

TABLE IV. Flange material.

Code	Flange material
C	CRES 304 (UNS S30400) or 316 (UNS S31600) in accordance with ASTM A276/A276M, ASTM A473, or SAE-AMS-QQ-S-763.
L	CRES 304L (UNS S30403) or 316L (UNS S31603) in accordance with ASTM A276/A276M, ASTM A473, or SAE-AMS-QQ-S-763.
N	Nickel-copper alloy in accordance with QQ-N-281, class A, or ASTM B127 (UNS N04400 only).

Vent requirement. The vent requirement shall be designated by the letter X, Y, or Z as indicated on figure 5 and figure 6. For flanges without a vent, the letter W shall be used. Machining for the vent port shall be as shown on the applicable figures.

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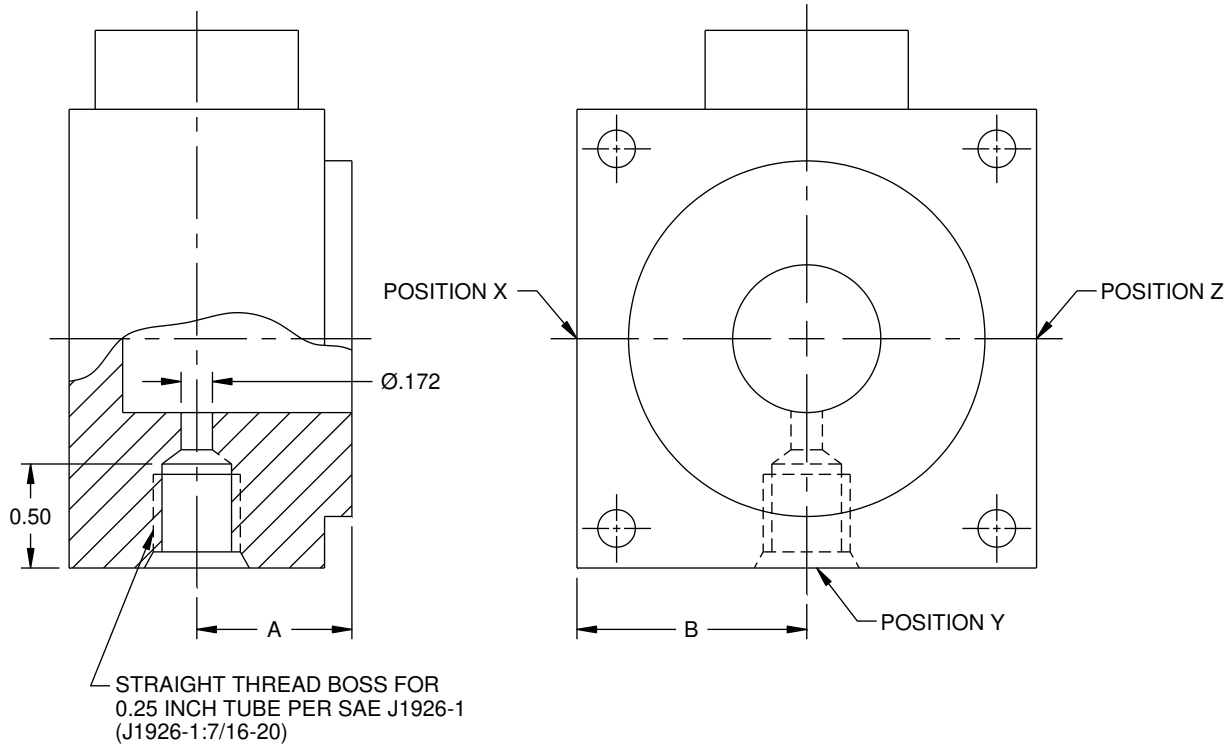
#	**	Flange size (inches)	Pipe size (NPS)	A	B
F	08	3.50	1.000	0.672	1.75
F	10	3.50	1.250	0.734	1.75
F	12	3.50	1.500	0.734	1.75
G	16	4.25	2.000	0.906	2.12

NOTES:

1. The vent port can be located at position X for the flanges in this figure. Insert the letter X at the appropriate location in the flange PIN. For flanges without a vent requirement, insert the letter W.
2. Unless otherwise specified herein or in 6.2 of the base specification (MIL-DTL-24704), tolerances shall be:
   a. X.XXX: ±0.015
   b. X.XX: ±0.03

FIGURE 5. Straight flange vent location.

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#	**	Flange size (inches)	Pipe size (NPS)	A	B
F	08	3.50	1.000	1.188	1.75
F	10	3.50	1.250	1.438	1.75
F	12	3.50	1.500	1.594	1.75
G	16	4.25	2.000	1.875	2.12

NOTES:

1. The vent port can be located at position X, Y, or Z for the flanges in this figure. Insert the letter X, Y, or Z at the appropriate location in the flange PIN. For flanges without a vent requirement, insert the letter W.
2. Unless otherwise specified herein or in 6.2 of the base specification (MIL-DTL-24704), tolerances shall be:
   a. X.XXX: ±0.015
   b. X.XX: ±0.03

FIGURE 6. Right angle flange vent location.

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CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 4810-2019-004)

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